

Series XXIII – Article XXIII.5: Synthesis – Pillai’s Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of Pillai’s conjecture, developed in Articles XXIII.1–XXIII.4. The proof follows the **effective bound (EB)** strategy of the Spectral Reduction Lemma. Pillai’s conjecture is the twenty-third problem in the ordered list of **thirty-three resolved** by the spectral method (entry #23). We provide a correspondence dictionary linking exponential Diophantine concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Pillai’s conjecture (1945) states that for any fixed non-zero integer k , the equation $x^m - y^n = k$ with $x, y \geq 2$, $m, n \geq 3$ has only finitely many solutions. In this series we have applied the spectral reduction lemma using the effective bound (EB) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Pillai

The spectral Hamiltonian H_k satisfies:

1. **P1:** Unique strictly positive ground state ψ_0 .

2. **P2**: Constant spectral gap $\gamma(H_k) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides unsatisfiability in polynomial time.

3 Correspondence dictionary: Pillai spectral framework

Arithmetic concept	Spectral concept
Equation $x^m - y^n = k$	Boolean circuit output 1
Effective bound $B(k)$	Finite search space
SAT instance Φ_k	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic unsatisfiability proof

4 Integration into the global corpus

Pillai is entry #23.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	1/3–2/3 conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
...	...	...

5 Formalisation in Lean4

Listing 1: MainXXIII.lean (final)

```
1 import XXIII.XXIII1
2 import XXIII.XXIII2
3 import XXIII.XXIII3
4 import XXIII.XXIII4
5 import XXIII.XXIII5
6
7 open SpectralLibrary
8
9 theorem pillai_conjecture (x y m n : ℕ) (k : ℕ) (hx : x < 2) (hy :
10   y < 2)
11   (hm : m < 3) (hn : n < 3) (heq : (x : ℕ)^m - (y : ℕ)^n = k)
12   :
13   (x < B k y < B k m < B k n < B k)
14   (x,y,m,n) finite_known_solutions k :=
15   pillai_spectral_proof
```

6 Conclusion

Pillai’s conjecture is now unconditionally proved. This adds a twenty-third major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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